THE NATIONAL UNIVERSITY OF LESOTHO

COURSE OUTLINE			
Course Title	Electronic Measurements and Instrumentation		
Course Code	EE 3400	**No. Of Credits**	4
Department	Physics and Electronics	**Faculty**	Science and Technology
Pre-requisites Course Code	P2508 (P238)	**Co- requisites Course Code**	P238, M2403
Course Coordinator(s)	Mr T. A. Koetje		
Email	t.koetje@nul.ls	**Telephone No.**	5221 3524 Ext. 3524
Other Course Teachers(s) Tutor(s)	Mr. S. Sehobai (Demonstrator)		
Learning Hours	Total: 60 Learning Hours Lecture (40 hrs), Laboratory (20 hrs)		
Student Quota	No limit		
Course Type	■ Core Course ☐ Compulsory ☐ Elective ☐ Other.....		
Offer in Academic Year	■ 1st Semester ☐ 2nd Semester ☐ Year Long		

COURSE DESCRIPTION

This course will introduce the fundamental concept of instrumentation and measurements, discuss fundamental theory and operating principles of instrumentation in industry. Covers the terminology, concepts, principles and computations needed to specify, analyse and maintain instrumentation systems. Design of various sensors will be investigated and the electronic signal-conditioning circuits required to convert the electrical changes in the transducers to signal which can be interpreted accurately by a microprocessor or embedded controller, are analysed and designed.

Students will learn concept of instrumentation and measurements, discuss fundamental theory and operating principles of instrumentation in industry. Practical examples will be used for common Filters, Pressure, Capacitive, Temperature, Flow, Motion and Motor & Actuator Systems.

COURSE OBJECTIVES

The course objectives are:

1. Understand the concept of a measurement and instrumentation systems to support accurate measurements
2. Understand the basic concepts in measurements (accuracy, precision, error,

conditioning, etc)

3. Understand operations of transducers, amplifier and signal processor
4. Be able to design variety of filters to control noise and frequencies
5. Be able to analyse and design various instrumentation sensors to perform different measurements

GUIDELINES ON GRADING AND STANDARDS OF ASSESSMENT

General Course Grade Descriptors

Please note: NUL has adapted standards-based assessment, thus, when using standards-based assessment as opposed to norm-based assessment, descriptors should be established so that students are clear about what is expected of them. Students will receive a score based on these benchmarks as opposed to ranking based on a norm.

A	Demonstrate evidence of original thought, strong analytical and critical abilities as well as a thorough grasp of the topic from background reading and analysis; Should demonstrate excellent organizational, rhetorical and presentational skills.
B	Demonstrate evidence of critical and analytical thinking but not necessarily original in their thinking; Show adequate grasp of the topic from background reading and analysis; should demonstrate strong organizational, rhetorical and presentational skills.
C	Demonstrate evidence of a reasonable grasp of the subject but most of the information is derivative, with rather little evidence of critical thinking; should demonstrate fair organizational, rhetorical and presentational skills.
D	Demonstrate evidence of being able to assemble the bare minimum of information, poorly digested and not very well organized in presentation. There is no evidence of critical thinking.
F	Demonstrate evidence of poor knowledge and understanding of the subject, a lack of coherence and organization, and answers are largely irrelevant. Work fails to reach degree level.
Non-graded	Inadequate attendance. However, the student has demonstrated evidence of a reasonable grasp of the subject and has achieved all major objectives of the course. (Please see and discuss more with the course coordinator)

COURSE CONTENT

Course topics include:

- Basic principles of measurements
- Characterisation and behaviour of typical measuring systems
- Data acquisition and Signal processing
- Resistance and Capacitance Measurement
- Torque and Strain Measurement
- Ranging Systems
- Passive and Active Filters
- Measurements of Temperature, Flow, Motion and Vibrations, Pressure and Force
- Motor Systems and Actuators

COURSE LEARNING OUTCOMES

Course Learning Outcomes	**Aligned Programme Learning outcomes**
After the course completion students would be able to	
1. Understand the general principles of measurement and instrumentation systems	1,2
2. Evaluate measurement systems by differentiating various characteristics (accuracy, precision, errors,etc)	1,2,3
3. Understand the design of various measurement sensor and techniques used to in their design	4,5
4. Understand the signals and noise, reliability of systems and be able to design measurement filters for different systems.	4
5. To perform practical experiments on design of measurement sensors and filters in the laboratory	1,2,3,4,5

TEACHING AND LEARNING APPROACH

There are four contact hours weekly, 2 x 1 hour lecture, 1 x 2 hour Labs / Tutorials

1. Lectures and presentation slides that provide theoretical background for electronic measurement and instrumentation in an interactive class discussions	ALL
2. Assignments and Tutorials that provide the opportunity for students to practise and apply the mathematical theories using particular problems and to understand the theory better	
3. Laboratories which include both simulation and hardware implementation of electronic sensors and filters	

COURSE ASSESSMENT METHODS

Assessment Method	**Description**	**Weight**	**Aligned Course Learning outcomes**

i. Assignments	To encourage group work and research, a maximum of three (3) assignments will be given to groups of three students.	5%	ALL
ii. Labs	A minimum of six experiments, covering the simulation and hardware implementation of material taught in class, are undertaken and lab reports submitted.	20%	ALL
iii. Tests	A minimum of three (3) invigilated tests will be given. The test covered topics discussed in class	25%	ALL
iv. Final exam	The end of semester exam covers all the material learned through class lectures, tutorials, laboratory sessions and assignments. The exam consists of five questions each of 25 Marks and the student will choose four questions to answer. The exam lasts for 3 hours.	50%	ALL

ESSENTIAL READINGS: (Journals, textbooks, website addresses, etc)

Textbooks:

1. Fundamentals of Instrumentation and Measurement by Dominique Placko, 2007
2. Fundamentals of Industrial Instrumentation and Process Control by Willam C. Dunn,2005
3. Any Electronics Measurement and Instrumentation books that can be helpful

MEANS/PROCESSES FOR STUDENT FEEDBACK ON COURSE

The student feedback questionnaire is one of the ways NUL courses and teaching are evaluated. NUL places significant importance on student learning and on continuous enhancement of teaching and learning outcomes.

Students are asked to complete this evaluation of their learning experiences at the conclusion of each course they enrol. Questionnaire items relate to the overall evaluation of the course as well as an evaluation of teaching.

COURSE POLICY (including plagiarism, academic honesty, attendance, etc)

Unless otherwise stated, the normal course administration policies and rules of the Faculty of Science and Technology apply. For the policy, see http://www.nul.ls
The University Regulations on academic misconduct will be strictly enforced. Please refer to http://www.nul.ls/student/plagiarism for further details. I addition, students should familiarise themselves with http://www.nul.lib.ls

General health and safety information can be obtained from http://www,nul.ls/health-and-safety

ADDITIONAL COURSE INFORMATION

The Course Coordinator is available for consultation at times that is displayed on the coordinator's office notice. Queries may also be emailed to the Course Coordinator directly at t.koetje@nul.ls

ACADEMIC COUNCIL APPROVAL

Signature: Date: